

1. GROUND REALITY: MORNING COOK DROPOUTS UNDER A 2-HOUR DELIVERY CLIFF

- 🔗**The 10:30 AM Crunch:** Home cooks prepare dal, sabzi, and fresh chapatis from scratch. When a cook drops out at 10:30 AM, backup kitchens require 90 minutes of active cooking before 12:15 PM delivery pickup. Ops has less than 15 minutes to reassign meals before lunch is physically unfulfillable.
- 🔗**The Unlogged WhatsApp Trap:** Cooks rarely update central spreadsheets. When Sunita Kulkarni (CK090) texted "Bhaiya aaj nahi ho payega" at 7:41 AM, ops got distracted in morning standup. The sheet remained "Active", and 6 Mumbai subscribers were left facing surprise hunger without notice.
- 🔗**The Jain Bottleneck Cascade:** Sequential manual matching causes severe allocation bias. An operator reassigning an early vegetarian order to Ayesha Gupta unknowingly burns her rare Jain-certified slot. Thirty minutes later, when a strict Jain customer arrives, no certified kitchen has capacity left.

2. TARGET PERSONAS & DAILY OPERATIONAL PRESSURES

- 🔗**City Operations Managers:** Managing daily dispatch across 500+ subscribers and 92 cooks. They operate under intense morning time pressure and need to resolve concurrent cook dropouts within 8 minutes with zero manual calculation errors.
- 🔗**City Business Heads (VP Ops):** Responsible for city unit economics and subscriber retention. They require 30-day visibility into repeat-offender cooks, chronic kitchen capacity deficits, and refund burn to restructure cook contracts and prevent churn before it happens.

3. MEASURABLE PRODUCTION GOALS (GROUNDED IN 7,138 ORDER HISTORY)

Goal Dimension	Specific Metric & Operational Formula	Current Baseline	Target SLA (v1.0)
North Star Outcome	Meal Recovery Rate = (Reassigned Meals Delivered / Total Impacted Orders) * 100	1.4% (Unmanaged)	>= 88.0% rescued
Resolution Velocity	Time-to-Resolution (TTR): Signal detection to batch notification dispatch	42 mins (Phone/Sheet)	<= 8 mins (Done by 10:45 AM)
Dietary Safety	Religious/Allergy Compliance: 0 tolerance for unverified Jain kitchen matching	3.2% Cross-contamination risk	0.0% (Hard Gate)
Financial Guardrail	Monthly Net Refund Burn: Cash refunds paid out due to cook cancellations	Rs 21,480 / month	<= Rs 12,000 / month (-44%)
Subscriber Retention	14-Day Post-Incident Churn: Subscribers cancelling after experiencing a no-show	28.5% Churn on unserved	<= 6.0% for rescued orders

4. WHAT THE PROTOTYPE PROVED (LIVE TESTING WITH SEED DATA)

- 🔗**Unstructured WhatsApp Signals Capture Reality:** Directly caught Sunita Kulkarni (CK090) via conversational Hinglish parsing, rescuing 6 meals in Mumbai that the operational spreadsheet missed entirely.
- 🔗**Global Batch Allocation Solves the Jain Bottleneck:** Solving all 24 morning orders simultaneously (Jain first -> Non-Veg -> Veg) recovered 5 additional meals compared to greedy sequential assignment.
- 🔗**Explainable Match Scoring Eliminates Operator Hesitation:** Displaying transparent criteria ("Bengaluru match | Jain certified kitchen | 12 spare slots | South Indian preference") cut operator review time to under 10 seconds.
- 🔗**Hard Kitchen Capacity Bounds Prevent Disaster:** Strictly respecting max_daily_orders prevented kitchen overloads, gracefully routing excess orders into automated UPI refunds + Rs 50 compensation.

5. DELIBERATE PRODUCT CUTS & ARCHITECTURAL TRADE-OFFS

- 🔗**No Two-Way Cook Negotiation Calls:** Calling cooks to ask "Can you take 4 more meals?" takes 25 minutes. In a 120-min window, this guarantees late delivery. Cut in favor of an opt-out standby agreement where cooks commit to absorb orders up to max_daily_orders.
- 🔗**No Subscriber Meal Choice Approvals:** Waiting for working professionals to approve substitute dishes during morning meetings takes >45 mins. Cut in favor of deterministic exact-diet auto-reassignment with instant proactive compensation credits.
- 🔗**No Dynamic Surge Bidding:** Surge pricing auctions encourage cooks to delay their availability to capture peak bonuses. Cut in favor of a predictable flat emergency bonus (+Rs 25/box) to protect supply trust and unit economics.
- 🔗**No Autonomous Zero-Click Execution:** Unverified WhatsApp texts contain sarcasm, retractions, and family members chatting. Cut full autonomy in favor of a 1-click human verification gate, preventing disastrous false cancellations.

6. OPEN ENGINEERING QUESTIONS BEFORE PRODUCTION BUILD

- 🔗**Concurrent Allocation Locking:** When two ops managers handle Bengaluru dropouts simultaneously, how do we implement row-level capacity locking (Redis mutex vs Postgres SELECT FOR UPDATE) to prevent double-booking a cook's remaining slots?
- 🔗**Kitchen Certification Standard:** What physical audit elevates a cook from "self-declared Jain" to "verified separate Jain kitchen" to eliminate religious liability under consumer safety standards?
- 🔗**Cook Response Falback Timer:** If an assigned backup cook does not acknowledge dispatch via WhatsApp within 12 minutes, what automated escalation triggers to re-route orders to standby cloud kitchens?
- 🔗**Meta WhatsApp Template Approvals:** Are pre-approved HSM utility message templates active for dynamic customer refund amounts, substitute cook names, and live dispatch tracking links?